# ASR-Aware Multi-Separator Fusion with Event-Conditioned Strength for Noise-Event Removal in Vaani Field Speech

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***Abstract.*** **Track 2 of the IndoML 2026 Datathon asks for noise events (breathing, horns, animals, alarms, music, machines, crying babies) to be removed from Vaani recordings while the speech stays intelligible. It is scored as SI-SDR on synthetic clips plus 100 times the drop in word error rate (ΔWER) of the frozen SraVaani-1.0 recogniser. We first established that this recogniser is hurt by speech distortion rather than by leftover noise, and built the system around that observation. Eighteen separators of three ClearerVoice-Studio architectures are fine-tuned on mixtures that reproduce the organisers' synthetic recipe. Their losses penalise artefacts more than residual noise and match the internal representation of the frozen recogniser; two of them are trained toward clean speech plus attenuated noise, and one subtracts an estimate of the noise. Four learned time-frequency fusion networks combine the separators, and noise-event time-stamps from our own Track 1 detector set the enhancement strength. One pipeline processes every clip. The final system scored a Combined 16.6113 (SI-SDR 13.82 dB, ΔWER +2.79 points, of which +7.85 on synthetic and +0.40 on natural clips) and placed eighth, as asymptoticchase.**

***Index Terms.*** **speech enhancement, noise event removal, ASR-aware training, model fusion, Vaani**

## I. Training Data and Preprocessing

**Task and metric.** Each of the 3,178 Phase 2 test clips must be returned as an enhanced 16 kHz waveform together with its SraVaani transcript. The score is the mean SI-SDR [1] on the synthetic clips, whose clean references are known, plus 100 times the pooled word error rate reduction over all clips. One point of ΔWER is therefore worth as much as one decibel.

**Vaani data.** The organisers released 695 synthetic validation pairs, each a noisy clip with its clean reference, together with 1,806 natural validation clips. We split the synthetic pairs with a fixed seed into 591 for training and 104 for checkpoint selection. From these pairs we measured the synthetic recipe. Noise occurs only inside one to four events of 0.2 to 1.1 s, placed at the times where the original recording's annotators marked noise. The per-event signal-to-noise ratio follows an empirical distribution with a median of −7.5 dB, so inside an event the noise is usually louder than the speech. The exact noise of each training pair is its noisy signal minus its clean reference, which gave 1,863 isolated noise events. For clean speech we added 3,968 spans from the Gold tier of the Vaani Noise-Event dataset [2], cut from the regions between annotated noise events of the verified-timestamp clips. No test audio was used for training.

**Additional datasets.** To widen the noise variety we used FSD50K [3], restricted per clip to CC0 and CC-BY licences, and the music-only `rfm` part of MUSAN [4] (CC BY 3.0). Clips were mapped to the seven organiser categories and cut into energetic segments of 0.2 to 1.2 s, giving 19,058 segments. The licence of every clip is listed in the repository. Clips from the FSD50K evaluation split, never used in training, were mixed into the 104 held-out clean references to form an unseen-noise selection set, and a copy of that set with every event 4 dB louder served as a hard low-SNR set.

**Augmentation pipeline.** Training mixtures are generated on the fly. A 3 s clean crop is scaled by a random gain between 0.5 and 1.5, and one to four events are placed at random non-overlapping positions. Each event is drawn from the organiser noise or, with probability 0.5, from the external bank, weighted by the organiser category frequencies. It is resampled to change its speed by up to 10% in either direction, its polarity is flipped with probability 0.5, and with probability 0.15 a second event is overlaid. The event is then scaled to an SNR drawn from the empirical list. For the members that emphasise low SNR, half of the events are lowered by up to a further 5 dB, because the Phase 2 low-SNR tail is 2 to 3 dB harder than the validation set. All audio is 16 kHz mono.

## II. Proposed Method

### A. Separators

All separators start from public ClearerVoice-Studio checkpoints [5] (Table I) and are fine-tuned in full (Fig. 1). Thirteen are MossFormerGAN-SE-16K generators [6], which predict a magnitude mask and a complex residual on a power-compressed STFT (window 400, hop 100). Three are MossFormer2-SE-48K mask networks [7] that run on audio resampled to 48 kHz, and two are MossFormer2-SS-16K time-domain separators whose first output was trained as speech. During inference each separator processes a clip in fixed windows (6 s, 3 s and 2 s for the three architectures) with 50% Hann overlap-add, and its output is scaled to the input level. The windows are the same for every clip; they recovered about 1 dB that models trained on 3 s crops lost on the 12 s Phase 2 clips.

**Losses.** For an estimate $y$, clean speech $c$ and noise $n$, we project $y$ onto the span of $c$ and $n$. The target part is $s_t = \langle y, c\rangle c / \|c\|^2$, the noise error $e_n$ is the remaining in-span part,

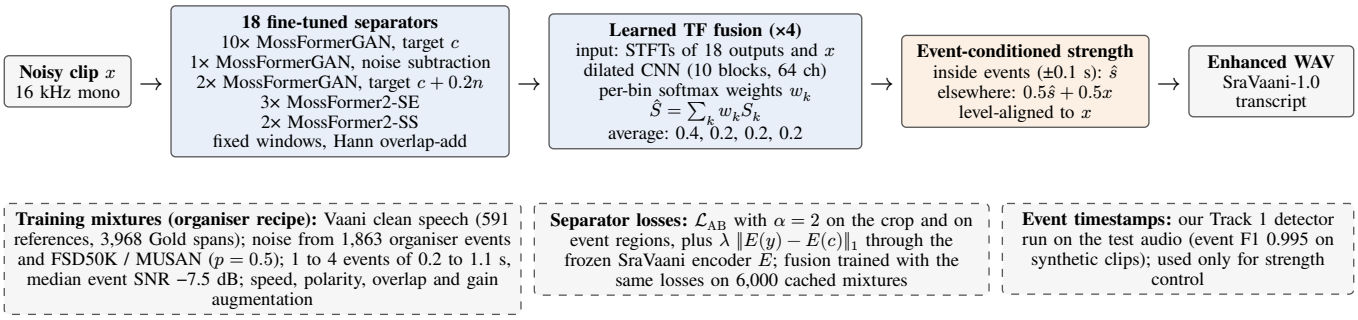



Fig. 1: Overview of the system. Every clip follows the same path; there is no natural/synthetic routing and no rule that decides whether to enhance.

and the artefact $e_a$ is the out-of-span residual. The artefact-boosted SI-SDR [8]

$$\mathcal{L}_{\mathrm{AB}} = -10 \log_{10} \frac{\|s_t\|^2}{\|e_n + \alpha e_a\|^2}, \quad \alpha = 2, \tag{1}$$

is applied to the whole crop and again to the event regions (noise support dilated by 0.1 s). For MossFormerGAN a compressed-magnitude MSE with weight 0.5 is added. We chose this loss after decomposing early outputs on the held-out pairs: clean speech plus our residual noise was recognised with a WER of 17.4, close to clean speech at 18.1, whereas clean speech plus our artefacts gave 23.9. The second loss uses the mandated recogniser. Its FastConformer encoder [9] is loaded frozen with its exact mel front end, and we add the L1 distance between its outputs for the enhanced and the clean signal on event frames, with weights from 1 to 10 across members. Gradients flow through the frozen encoder into the separator.

**Target variants.** One MossFormerGAN predicts the noise instead of the speech: its decoders output a mask and residual for the noise, which are subtracted from the input in the compressed complex domain. Its output heads start at zero, so it begins as the identity and leaves speech untouched unless it learns otherwise. Two further members are trained toward a partial-noise target $t = c + 0.2n$. Because $x - t = 0.8n$, all losses apply unchanged. These models are not forced to remove the last trace of the noise, which is what makes a separator distort speech.

**Training strategy.** Each member is fine-tuned with AdamW (weight decay $10^{-4}$, cosine decay to 1/20) at a learning rate between $10^{-5}$ and $5 \cdot 10^{-5}$, batch 2, for 3,000 to 8,000 steps. Each starts either from the pretrained weights or from an earlier in-domain member. Diversity comes from the ASR-loss weight, the seed, the low-SNR emphasis, training on all 695 pairs, 12 s crops, and the three architectures and targets. Checkpoints are selected on the held-out and unseen-noise sets.

TABLE I: Pretrained models and checkpoints used by the final system

| Name (version) | Source | Role |
|---|---|---|
| MossFormerGAN-SE-16K (rev. dcb1b20) | HF alibabasglab [6] | init. of 13 separators |
| MossFormer2-SE-48K (rev. eff8c97) | HF alibabasglab [7] | init. of 3 separators |
| MossFormer2-SS-16K (rev. 407cb03) | HF alibabasglab [7] | init. of 2 separators |
| ClearerVoice code (clearvoice 0.1.2) | ModelScope GitHub [5] | model architectures |
| SraVaani-1.0 (rev. f5dd535) | HF ARTPARK-IISc [10] | transcription; frozen loss encoder |
| Track 1 detector (ours) | our Track 1 release [11] | noise-event timestamps |
| 18 separators, 4 fusions (ours) | this work (`CHECKPOINTS.md`) | final system |

### *B. Learned Fusion*

Averaging separators of one architecture saturated quickly, because their artefacts are correlated, so we learn the combination. The fusion network takes $\log(1 + 10\,|S_k|)$ of the STFTs (512 points, hop 128) of the $K$ separator outputs and of the noisy input. It applies a 3×3 convolution to 64 channels and ten residual blocks of dilated (1×d, d = 1, 2, 4, 8, 16, twice) convolutions with group normalisation and PReLU. A final 1×1 convolution gives $K + 1$ logits per time-frequency bin, and their softmax weights $w_k$ form the output $\sum_k w_k S_k$, which is inverted and scaled to the input level. The network has 0.38 M parameters. Its last layer starts at zero, with a bias that encodes a fixed blend, so training begins from that blend. The fusion is trained on 3,000 to 6,000 cached recipe mixtures run once through every separator, with the AB-SI-SDR and SraVaani losses (AdamW, batch 8, 4,000 to 6,000 steps). The submitted output averages four such networks over 18, 16, 10 and 10 separators with weights 0.4, 0.2, 0.2 and 0.2.

### *C. Event-Conditioned Strength*

Noise-event timestamps come from our Track 1 system [11], run on the test audio; its event F1 on synthetic test clips is 0.995. Inside each detected event, widened by 0.1 s with 10 ms ramps, the output is the fused signal. Elsewhere it is half fused signal and half input, after a least-squares level match, so model output is present on every sample. A clip with no detected event receives the full fused output. There is no noise dictionary, no correlation search and no rule that decides whether to enhance. The output is peak-limited to 0.99 and written as 16-bit PCM.

**Implementation.** Python 3.12, PyTorch 2.13 (CUDA 13.0), on one NVIDIA RTX PRO 6000 (96 GB) or A100 (80 GB) per run. A separator takes 2 to 4 GPU-hours to fine-tune and a fusion network under one hour. Inference needs 4.7 GB of GPU memory and about 14 s per 12 s clip.

## III. Results and Conclusion

Table II shows how the Phase 2 score developed; each row was a separate submission. The final system scored a Combined **16.6113**: SI-SDR 13.8237 dB on the synthetic clips and ΔWER +2.7876 points on all clips, of which +7.8507 on synthetic and +0.4012 on natural clips. It placed **eighth** under the name asymptoticchase. Rerun from raw audio with the released code, it reproduces the submitted waveforms to within 67 to 74 dB, with identical transcripts.

TABLE II: Phase 2 test results; each row adds one component

| System (Phase 2 test) | SI-SDR | ΔWER syn. | Comb. |
|---|---|---|---|
| Earlier BiGRU masking network | 7.74 | −1.57 | 7.06 |
| In-domain MossFormerGAN and MossFormer2 | 12.97 | 5.68 | 14.98 |
| + windows, ASR loss, averaging, MF2-SS | 12.96 | 6.97 | 15.41 |
| + learned fusion (6 separators) | 13.47 | 7.51 | 16.14 |
| + fusion over 10 and 16 separators | 13.77 | 7.85 | 16.56 |
| + partial-noise separators (18) | 13.84 | 7.73 | 16.58 |
| **Average of four fusions (final)** | **13.82** | **7.85** | **16.61** |

The largest single gains came from in-domain fine-tuning (+7.9 over our earlier masking network) and from the learned fusion (+0.73). The SraVaani feature loss added 0.22, almost all of it in ΔWER. On the 104 held-out pairs the system lowers the WER of the noisy input from 30.2 to 23.5 and raises SI-SDR from −0.8 to 17.8 dB, while the WER of natural clips stays unchanged (24.2 to 24.3).

Several measurements bound what is left. If our output is kept outside the events and clean speech is inserted inside them, the WER falls to 17.7, the level of clean speech; with our output inside and clean speech outside it stays at 23.8. All remaining recognition errors therefore come from separation inside events at −7.5 to −15 dB, where we delete twice as many words as an oracle ideal ratio mask (45 against 22), which itself reaches clean-level WER. The real SI-SDR trails the held-out value by 4 dB. About half of the Phase 2 noise has no close match in training (normalised cross-correlation above 0.5 for 48% of segments, against 100% on held-out pairs), unseen speakers cost about 0.9 dB, and the test has a harder low-SNR tail. Several ideas did not help. Feeding the event mask into the separator gave the same SI-SDR with detector, perfect or no timestamps (17.00, 17.00, 16.95). Choosing among our outputs per clip by consensus, recogniser confidence or a trained selector lowered WER by at most 0.8 points while costing more SI-SDR. Fine-tuning the URGENT 2025 TF-GridNet baseline [12], [13] stayed below MossFormerGAN, and neither external noise nor 1,039 noise events extracted from Gold clips added anything inside the fusion.

In conclusion, the mandated recogniser tolerates residual noise but not speech distortion. Losses that weight artefacts and follow the recogniser's own features, separators with less aggressive targets, and a learned fusion across architectures gave our improvements, and event timestamps were most useful for deciding where to process. The remaining gap is separation quality at very low in-event SNR on unseen noise, which larger models trained on more diverse in-domain noise would address. Code, configurations and checkpoints are released with this report.

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